BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI

Work Integrated Learning Programmes Division

AIML ZG535 — Machine Learning on Edge · Mini-Project (EC-I)

LogiEdge — Phase 1 Report

System Architecture & Hardware Justification (Modules 1–2, Components A & B) — Group 9

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Task A1 — Constraint Analysis

Latency. A refrigeration failure raises cargo temperature by 1 °C/minute, and the 90-second detection SLA allows at most a 1.5 °C rise before an alert must fire. Cloud inference requires a round trip over rural cellular infrastructure that, on the Nashik–Aurangabad route alone, loses connectivity entirely for 35–90 minutes at seven documented locations. During any such gap a cloud-only system cannot receive sensor data or transmit an alert at all — an outage of up to 90 minutes is sixty times the SLA, and precisely the failure mode behind the ₹28 lakh vaccine spoilage incident. On-device TFLite inference completes in well under a millisecond, so the SLA is trivially met by an edge architecture and structurally unmeetable by a cloud-dependent one.

Bandwidth. At 1 Hz temperature and 500 Hz 3-axis vibration sampling, a single truck generates roughly 519 MB of raw sensor data per day, costing approximately ₹52/truck/day (about ₹16.1 lakh/year for the 85-truck pilot) at ₹0.10/MB. Processing at the edge and transmitting only classification results — one ~110-byte message every 10 seconds — reduces this to about 0.95 MB/truck/day, roughly 546 times less data and under ₹3,000/year for the whole pilot fleet. Edge processing here is not an optimisation; it is the difference between a viable and a non-viable telecom bill.

Connectivity. During the documented 35–90 minute signal gaps a cloud-only system is blind: no ingestion, no inference, no alerting, and no chain-of-custody records for the very interval where an undetected failure is most damaging. The LogiEdge edge node classifies every window locally regardless of connectivity, raises the driver alert immediately, and appends every Warning/Critical result to a durable local alert log; once coverage returns, unsynced records are automatically republished to the operations centre. This store-and-forward behaviour was verified by seeding an unsynced alert-log entry with the broker unreachable and confirming automatic republication on reconnect.

Privacy. Because inference runs on the truck, raw pharmaceutical cargo telemetry never leaves the vehicle — only a class label, a confidence score, and a timestamp are transmitted, over an authenticated TLS username/password MQTT channel in the production broker profile, with per-role access-control lists (sensor publisher, inference consumer, fleet-ops monitor). This lets FreightBridge contractually demonstrate to its pharmaceutical clients that condition data cannot be accessed by unauthorised third parties: the raw data never transits a third-party network at all, and the little that does transit is role-scoped, encrypted, and auditable.

Task A2 — System Architecture

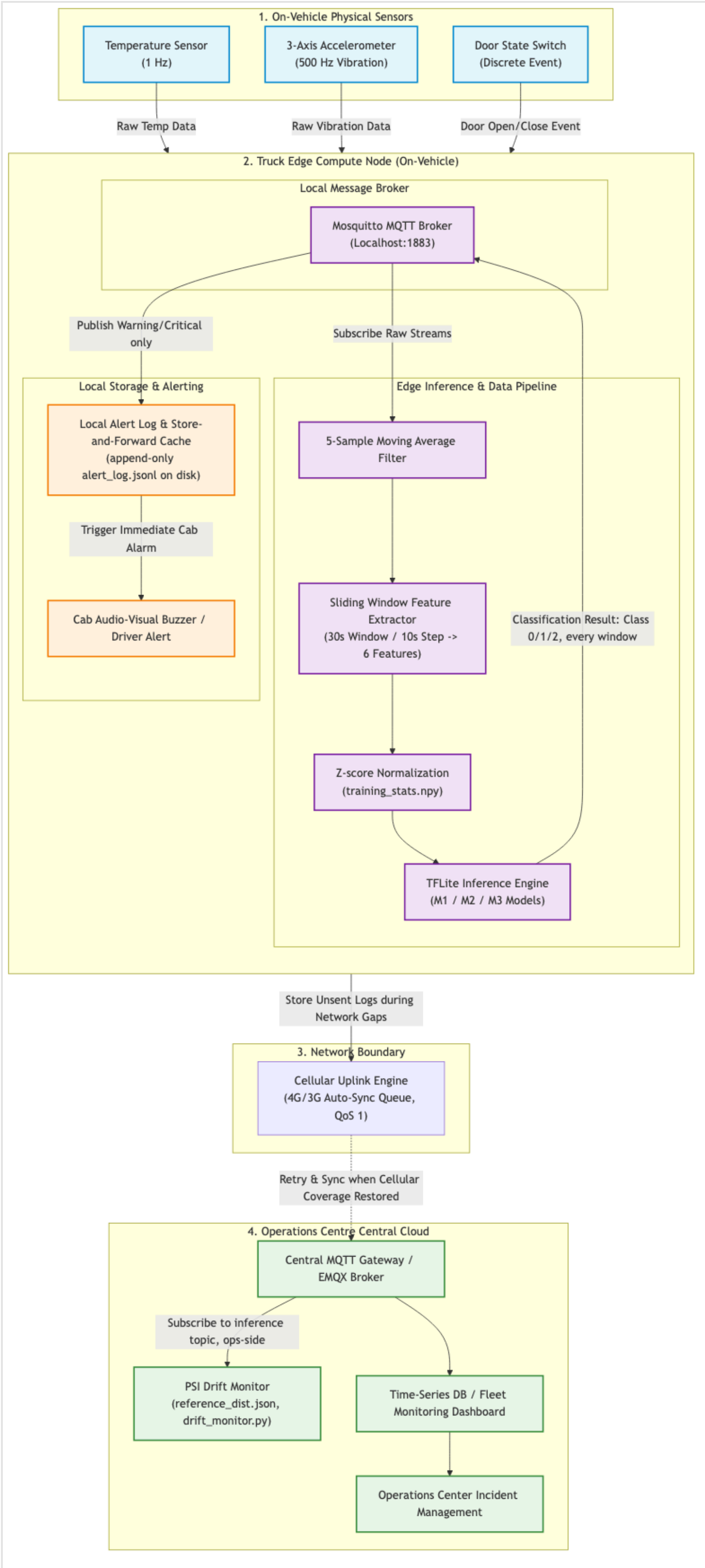

Figure 1. LogiEdge system architecture: on-vehicle sensors, local Mosquitto broker, preprocessing + TFLite inference pipeline, durable local alert log with driver alarm, cellular uplink with store-and-forward sync, and operations-centre backend.

Task B1 — Constraint Triangle Application

Option	Performance	Power	Cost (85 / 265 trucks)	Verdict
Raspberry Pi 5 + AI HAT+ (13 TOPS)	Ample for an 803-parameter MLP; runs full Docker/MQTT/Ansible stack	7.5 W — 25% headroom under the 10 W budget	₹12.75 L / ₹39.75 L	**Selected**
Jetson Orin Nano Super (67 TOPS)	5× the compute this workload can use	15 W — exceeds the 10 W budget	₹38.25 L / ₹1.19 Cr	Rejected
STM32H7 custom MCU	Cannot run the Docker/MQTT/Ansible MLOps stack	0.4 W — excellent	₹2.98 L / ₹9.28 L	Rejected

The dominant Constraint Triangle vertex for this deployment is **power-qualified lifecycle cost under a hard reliability constraint** — not peak performance. The Pi 5 + AI HAT+ meets the 90-second SLA with sub-millisecond inference, sits inside the 10 W budget with 25% headroom, and costs a third of the Jetson at full 265-truck scale for compute this model does not need. The Jetson's extra 67 TOPS buys nothing for a 6-feature MLP while breaking the power budget from the 12 V truck supply. The STM32H7 saves capital cost but forfeits the Docker containerisation, OTA update path, and drift-monitoring capability the pilot exists to prove — a false economy for a safety-critical fleet deployment.

Task B2 — Arithmetic Intensity and Roofline Analysis

For the given workload of 45 MFLOPs and 18 MB accessed per inference, on the Pi 5 CPU (16 GFLOP/s NEON SIMD, 12 GB/s LPDDR4X):

Quantity	Value
Arithmetic Intensity = 45 MFLOP / 18 MB	**2.5 FLOP/byte**
Ridge point = 16 GFLOP/s / 12 GB/s	**1.33 FLOP/byte**
Classification (2.5 > 1.33)	**Compute-bound**

The model sits to the right of the ridge point, so it is **compute-bound**: additional memory bandwidth would not reduce latency. The optimisation lever that will is reducing arithmetic per inference — INT8 quantisation (NEON-accelerated integer maths) and structured pruning to shrink FLOPs — which is exactly the optimisation programme implemented in the Phase 2/Final work (Component F).